

By application, the market can be categorised as:

- Diagnosis
- Monitoring
- Treatment

By component type, it can be categorised as:

- Sensors
- Batteries
- Memory devices
- Displays
- Microprocessors/microcontrollers

Opportunities galore

The medical electronics industry has achieved double-digit growth in recent years, and this growth trajectory is expected to continue owing to the rising incidence of chronic diseases, increased urbanisation, growing elderly population and increased awareness about latest technology solutions to save, augment and rehabilitate life.

Semiconductors play an important role in shaping the medical electronics device market. This market may, in turn, represent a significant growth opportunity for semiconductor players. Performance of medical electronics relies on innovations from semiconductor players—highly-integrated microchip designs. Use of semiconductors is critical in power supplies to displays in medical equipment. Semiconductors are an integral part of most medical equipment, starting from high-end imaging to small hand-held devices. Role of semiconductor suppliers becomes more critical as medical equipment become portable and smaller.

Moreover, there is a growing demand for lower cost systems and more sophisticated hospital equip-

IIPME: An interesting policy boost

The Ministry of Electronics and Information Technology (MeitY) and Biotechnology Industry Research Assistance Council (BIRAC), Department of Biotechnology, Ministry of Science and Technology, government of India, has launched a collaborative project, called Industry Innovation Programme on Medical Electronics (IIPME). The project goal is to fund a portfolio of Indian-led pilot projects targeting innovations in multidisciplinary areas comprising electronics, engineering, medical devices, healthcare, software, algorithms and information technology, to help the medical electronics fraternity and to bring in fast-pace research and development in this area.

The idea is to provide funding support to applicants for testing their bold ideas, mentorship from various subject matter experts, networking platforms and an opportunity to scale up their technology. Under this project, support (grants) will be provided at seed, early transition and transition-to-scale stages.

Seed. This grant is for funding projects that are at the initial stages of the product development cycle. It does not require preliminary data and is meant to provide an opportunity to test particularly bold ideas. A grant of up to ₹ 5 million for a period of 18 months can be provided in this category.

Early transition. This category is for projects that have established proof of concept (POC) and require incremental prototype innovations and validation. It is for small-risk projects that are at the early stages of translational research. A mix of grant and loan not exceeding ₹ 10 million over a period of 24 months can be provided in this category.

Transition-to-scale. This stage is for projects that have already shown promising data on establishing POC and have generated enough validation data. It requires demonstration of detailed preliminary data, and is meant to provide an opportunity to develop, refine and rigorously test approaches that have previously shown promise in controlled or limited settings. A mix of grant and loans for a period of 24 to 36 months is proposed. The project cost is matched equally by government funds and the industry.

(Source: www.birac.nic.in)

ment in the Indian market. Significant growth in lower cost wearable medical systems that wirelessly transmit patients' vital signs to doctors via cellphones or the Internet and fitness monitors that are used as activity trackers has led to significant demand for smarter and more efficient semiconductors.

Market trends

The therapeutic segment in the medical electronics market is expected to witness significant growth in the next five years. With increasing awareness about cardiovascular and neurological issues worldwide, demand for therapeutic devices such as pacemakers, implantable cardioverter defibrillators and other devices is going to increase.

The diagnostic segment captured considerable revenue share in 2018 and, according to Global Market Insight, it is expected to achieve 13.1 per cent growth in the next five years. Increasing awareness about the benefits of early diagnosis is going to trigger the demand for better and accurate diagnostic medical electronics devices. Further, growing occurrence of genetic diseases will propel the demand for relevant diagnostic devices, augmenting segmental growth.

The hospital segment, specifically with huge demand for medical electronics devices related to cardiovascular diseases and respiratory infections among geriatric population, accounted for 45.5 per cent of the global medical electronics market in 2018. The growth is expected to continue and is anticipated to have significant revenue share over the next five years. Moreover, the need for a well-equipped public healthcare system would propel the segmental growth further.

Additionally, governments in developing countries including India implement certain initiatives that ensures availability of superior-quality medical electronics devices, thereby, fostering regional growth in forthcoming years. **EFY**

SEMICONDUCTOR COMPONENTS USED IN THE MEDICAL ELECTRONICS SECTOR

Product/segment	Widely-used semiconductor components
Medical imaging	Microcontrollers, microprocessors, amplifiers, data converters, digital signal processors, application-specific standard products and application-specific integrated circuits
Patient monitoring systems	Microcontrollers, sensors, ZigBee modules, application-specific integrated circuits, rectifiers, amplifiers, controllers, EEPROMs
Digital hearing aids	Microcontrollers, EEPROMs, linear accelerometers, comparators, amplifiers, audio processor integrated circuits and other application-specific integrated circuits
Infusion pumps	Microcontrollers, regulators, transistors, rectifiers, comparators, amplifiers, sensors